

Terms

1 Global warming hiatus

The period roughly 1998–2012 during which the rate of global mean surface temperature increase slowed relative to the preceding and following decades. Attributed to a combination of ocean heat uptake, internal variability (PDO negative phase), volcanic aerosols, and incomplete radiative forcing accounting (Medhaug et al. 2017, Nature).

Not a pause in warming — total heat content continued to rise — but a slowdown in the rate of surface temperature increase.

2 Regime shift

An abrupt, persistent change in the statistical properties (typically the mean) of a time series. Detected by algorithms such as STARS. Not the same as a trend change; a regime shift implies a relatively sudden transition between two quasi-stable states.

3 Warming

In the context of lake temperature: the total change in LSWT over a study period, typically estimated via Theil–Sen slope $\times$ period length.

4 Acceleration

The rate of change of the warming trend. Positive acceleration means warming is speeding up; negative means warming is slowing (not cooling). Estimated via Sen slope of first-differenced STL trend.